UPDATE BRIEF FOR *AN ORDER ON THE LEECH LATTICE FROM A $\mathbb{Z}_3$-SYMMETRIC TRIPLE-OCTONION PRODUCT* V4 (FROZEN 25 MAY 2026)

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Abstract. The main paper is frozen at v4 (25 May 2026); the working source is `paper/main.tex` and the frozen snapshot is `paper/main_2026-05-25.tex`. This brief records two items for a future revision: (1) the Petersson 2018 reference is presently dangling in the bibliography and should be cited in the body text; (2) the triple-octonion construction is in fact S_3-symmetric on the three blocks, not merely $\mathbb{Z}_3$-symmetric. Both items are documented here against v4 and are not yet folded into the manuscript.

1. Petersson 2018 – dangling reference

Status. The reference `Petersson2018` appears in the v4 bibliography (H. P. Petersson, "Integral octonions," lecture at the Málaga Workshop on Non-Associative Algebras, 2018) but is not cited anywhere in the body of v4. This is a regression from earlier versions of the paper, where Petersson's lecture was cited as the key modern account of the integral-octonion history.

Why this is important. Petersson's 2018 lecture is the link in the historical chain that brings the 1923–1946 narrative into the present:

- It is the source from which this project first identified Kirmse's J_1 as a specific non-closed candidate (and not the whole of Kirmse's work).
- It treats Kirmse with appropriate care – calling his work "the bold attempt of a young mathematician" – and is therefore the reference for the fair-attribution stance the present paper takes in Appendix B.
- It also flags Dickson's 1923 priority, which the paper now records in Appendix B's opening.

Proposed insertions for the next revision.
End of paragraph in §2.2 (*The E_8 lattice*). After the existing sentence ending " closed under multiplication and admits no enlargement to a larger order" (citing Coxeter 1946 and Conway–Smith 2003), append a clause crediting Petersson:

> "...; for a modern survey of the integral-octonion history, see Petersson (2018)."

Appendix B.. Petersson 2018 is the natural citation in the historical note where the project's stance on Kirmse is established. Two natural spots:

- In the **Kirmse (1924)** paragraph, after "The remaining seven of his eight are the genuine maximal orders." add a parenthetical: "A modern account of Kirmse's contribution, treating it as the bold work of a young mathematician on a topic weird and bizarre even today, is in Petersson (2018)."
- Alternatively, at the start of Appendix B, in the opening sentence "The construction this paper builds on was developed across four primary papers between 1923 and 1946," append a pointer to Petersson (2018) as the modern source from which this project first traced the chain.

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Either placement (or both) closes the dangling reference and restores the credit that earlier versions had.

2. Kirmse 1924 – two further contributions worth crediting

The author's reading of Kirmse 1924 in the original highlights two further contributions that the v4 Appendix B Kirmse paragraph does not currently capture and that deserve credit.
(a) A nascent ideal theory on the E_8 lattice. Kirmse 1924 develops an ideal-theoretic treatment of the integral octonions, working out divisibility and ideal-type structure on the lattice he has constructed. This is an early and substantive companion contribution to the multiplicative-closure question that the v4 Kirmse paragraph focuses on – and it is the first such development on an E_8-type lattice of integral octonions. Mahler 1942 takes up the ideal-theoretic line again with the Euclidean property and the principal-ideal theorem, but Kirmse's earlier work in this direction is currently underrepresented.
(b) Algebra alternativity as a foundational building block. Kirmse takes *alternativity* – the weakened associativity available in the octonions (and more generally, in alternative algebras) – as a basic axiom on which to build his treatment, rather than working from full associativity and recovering alternativity as a consequence. This is a methodological choice that deserves recognition in its own right: it suggests a programme of replacing associativity by alternativity at the foundation of an arithmetic of integral octonions, and it anticipates the structural role alternativity plays in Cayley–Dickson algebras and their integral arithmetic in the later literature.

Proposed handling. In a future revision, extend the v4 Appendix B Kirmse paragraph with one or two sentences acknowledging both contributions. A draft:

> "Kirmse also develops an ideal-theoretic treatment of the integral octonions – a substantive companion contribution that is taken up again in Mahler (1942) – and methodically takes algebra alternativity as a foundational constraint, anticipating the structural role alternativity plays in later integral-octonion arithmetic."

This complements the Petersson 2018 citation proposed in §1: the Petersson reference establishes the project's fair-attribution stance toward Kirmse, and these two sentences make that stance concrete by naming the specific further contributions that the narrowly-focused v4 paragraph leaves to the side.

3. S_3 symmetry on the three blocks

Finding. The triple-octonion product $\star$ (Definition 3.3 of v4) is in fact S_3-symmetric on the three blocks: every permutation in S_3 of the indices $\{1, 2, 3\}$ is an automorphism of $(\Lambda, +, \star)$.

Verification. The two generators of S_3 are tested in `python_project/src/probe_aut_lambda_star.py` using the exact $\mathbb{Z}$-basis of Λ from `verify_consecutive_twists_exact.py`; for each candidate g on the 24-vector basis B of Λ, the script checks $g(B_i \star B_j) = g(B_i) \star g(B_j)$ for all $24 \times 24 = 576$ basis pairs.

candidate	verdict
$\mathbb{Z}_3$ cyclic block permutation $(x, y, z) \mapsto (y, z, x)$	preserves Λ AND $\star$
S_3 transposition of blocks $1 \leftrightarrow 2$	preserves Λ AND $\star$
componentwise negation $-I_{24}$	preserves Λ, does NOT preserve $\star$
σ applied to each block	does NOT preserve Λ

The first two candidates generate S_3: a 3-cycle and a transposition on $\{1, 2, 3\}$ together generate the full S_3. Hence every S_3 permutation of the three octonion blocks is in $\mathrm{Aut}(\Lambda, +, \star)$.

Why this is more than "$\mathbb{Z}_3$-symmetric". The paper's title and the definition of $\star$ both emphasise the $\mathbb{Z}_3$ routing: same multiplication in all three blocks, with a cyclic $\mathbb{Z}_3$ routing of cross-block products. But $\mathbb{Z}_3$ is only the cyclic group of order three; the empirical finding here is that the construction is symmetric under the full S_3, which is generated by the cyclic $\mathbb{Z}_3$ plus any single block-transposition.

In other words, the $\mathbb{Z}_3$ routing on the right-hand side of Definition 3.3 is consistent with a stronger S_3 symmetry of the resulting bilinear product on $\mathbb{R}^{24}$. This is empirically visible only by checking a block-transposition explicitly (a $\mathbb{Z}_3$-symmetry argument from the definition alone gives only the 3-cycle).

Bonus: strict subgroup of Co_0. The same probe also records:

- $-I_{24} \in \mathrm{Co}_0 \setminus \mathrm{Aut}(\Lambda, +, \star)$: the componentwise negation lies in the Conway group Co_0 (preserves Λ), but does not preserve $\star$ (since $\star$ is bilinear, $-I_{24}(u) \star -I_{24}(v) = u \star v$, whereas $-I_{24}$ acting on the result of $u \star v$ would give $-(u \star v)$; the two disagree unless $u \star v = 0$).
- Therefore $\mathrm{Aut}(\Lambda, +, \star) \subsetneq \mathrm{Co}_0$ is a *strict* inclusion.

This is consistent with the paper's §7 open question on the automorphism group of $(\Lambda, +, \star)$: the question is still genuinely open, but the bound from below has improved (a copy of S_3 inside Aut rather than only $\mathbb{Z}_3$) and the bound from above has improved (strict inclusion $\mathrm{Aut}(\Lambda, +, \star) \subsetneq \mathrm{Co}_0$).

Proposed handling.

(1) Note the S_3 symmetry in v4's §6 "**Comparison.**" paragraph, which currently says "the cross-block $\mathbb{Z}_3$ symmetry is supplied by the routing of Definition 3.3 rather than by the octonion product itself" – replace "$\mathbb{Z}_3$" with "S_3" (or "S_3 extending the $\mathbb{Z}_3$ routing").

(2) Alternatively (or in addition), add a one-line remark at the end of §3 noting that the routing in Definition 3.3 induces an S_3 symmetry on the three blocks, verified empirically.

(3) Update §7's bullet on the automorphism group to record $\mathrm{Aut}(\Lambda, +, \star) \supseteq S_3$ (block permutations) and $\mathrm{Aut}(\Lambda, +, \star) \subsetneq \mathrm{Co}_0$ (strict inclusion from $-I_{24}$).

No retitling is needed: the paper's title "$\mathbb{Z}_3$-symmetric triple-octonion product" refers to the $\mathbb{Z}_3$ routing rule used to *define* the product, not to the full symmetry of the resulting object. $\mathbb{Z}_3$ in the title remains accurate as a description of the construction.

Cross-references

- Main paper: `paper/main.tex` (v4, frozen 25 May 2026 at `paper/main_2026-05-25.tex`).
- v3 → v4 changelog: `paper/v3_to_v4_summary.md`.
- S_3 probe script: `python_project/src/probe_aut_lambda_star.py`.
- Petersson 2018 (URL preserved in v4 bibliography): `https://www.fernuni-hagen.de/mi/fakultaet/emer`